MD. SAZZAD HOSSAIN

ID: 5563395853 **Passport:** A20668458 **Date of birth:** 25/06/2003 **Phone:** (+880) 01303337052 (Whatsapp) **Email address:** md.sazzad.eee@gmail.com **LinkedIn:** MD. SAZZAD HOSSAIN **Website:** https://sazzad-eee.github.io/Portfolio/

Address: Banani, Block-B, Road 23/A, House 09, 1213, Dhaka, Bangladesh (Home)

ABOUT MYSELF

I am Md. Sazzad Hossain, an EEE graduate with versatile skills across practical electrical work and modern engineering tools. My future goal is to grow as a well rounded professional who can adapt to new challenges, learn continuously, and contribute to meaningful solutions.

EDUCATION & TRAINING

Bachelor of Science in Electrical and Electronic Engineering

American International University-Bangladesh [23/01/2022 - 26/12/2025]

City: Dhaka | **Country:** Bangladesh

Website: https://www.aiub.edu/ | **Field(s) of study:** Engineering: | **Final grade:** 3.57/4.00 | **Number of credits:** 148

Higher Secondary Certificate (HSC)

Qadirabad Cantonment Sapper College [01/01/2018 - 30/01/2021]

City: Natore | **Country:** Bangladesh

Website: https://qcsc.edu.bd/ | **Field(s) of study:** Science: | **Final grade:** 4.67/5.00

Secondary School Certificate (SSC)

Jigori High School [01/01/2016 - 06/05/2018]

City: Natore | **Country:** Bangladesh

Website: https://jigorihighschool.com/en/ | **Field(s) of study:** Science: | **Final grade:** 4.89/5.00

PUBLICATIONS

[2026]

A Physics-Informed Self-Calibrating Digital Twin with Conformal Uncertainty for Multi-Inverter

Status: Under Review

- Developed a novel physics-informed digital twin framework integrating residual learning, conformal uncertainty quantification, and online self-calibration for multi-inverter photovoltaic systems. Demonstrated improved prediction accuracy and inverter-level anomaly detection using real-world PV plant data.

Journal Name: TENSYMP

WORK EXPERIENCE

Energypac Engineering Ltd. - Bangladesh, Dhaka

Website: https://www.energypac.com/epgl/

Internship Trainee

[28/06/2025 - 21/07/2025]

- Gained hands-on experience in manufacturing and testing of power equipment including transformers, switchgear, and protective devices

- Assisted with equipment assembly, relay settings, and quality inspections
- Participated in high voltage testing procedures in accordance with IEC standards
- Supported testing and commissioning activities for power distribution equipment

LANGUAGE SKILLS

Mother tongue(s): Bengali

Other language(s): Hindi

English

LISTENING: C1 **READING:** B2 **WRITING:** B2
SPOKEN PRODUCTION: C1
SPOKEN INTERACTION: C1

SKILLS

Hardware Design, Circuits & PCB

EasyEDA | NI Multisim | Proteus

EDA, IC & CAD Tools

Cadence Virtuoso | AutoCAD | AutoCAD Electrical | SOLIDWORKS

Simulation, RF/EM & Optics

CST Microwave Studio | Antenna Design | OptiSystem

Software, Programming & Data Tools

C | C++ | Python | NumPy | Pandas | SciPy | Matplotlib | Seaborn | TensorFlow | PyTorch | Scikit-learn | Git/GitHub | OpenCV | MATLB and Simulink

Energy Systems

Renewable Energy | HOMER Energy | RETScreen

Professional Skills

Research Skills | Problem Solving | Team Leadership | Project Management | Presentations | Marketing

Embedded Systems

Raspberry Pi 5 | ESP32 | Arduino Uno | AI vision kit | Motion sensors

PROJECTS

[07/04/2024 - 23/05/2025]

- Designed an intelligent edge computing system that captures and processes images using computer vision algorithms to detect motion through sequential frame analysis
- Implemented AI-based classification on captured images to identify motion events and trigger automated responses
- Integrated AI Camera Module with Raspberry Pi 5 and Vision Kit for complete on-device processing without cloud dependency
- Optimized image capture pipeline and motion detection algorithms for low-latency performance on edge hardware